

range of 10ns to 20ns, to an equivalent electrical signal.

The peak power of the received laser pulse could be as low as a few tens of nano-watts when ranging a far-off target and as high as a few tens of milli-watts when the target is close by. This implies that the amplifier portion of the front-end needs to have a dynamic range as high as 100dB to 110dB. This is usually achieved partly in the avalanche photodiode by controlling the responsivity of the device through its reverse-bias variation and partly in the gain-controlled amplifier stage.

Laser pulse width decides the bandwidth of the front-end and is given by $\text{bandwidth} = 350/t_r$ where t_r is rise time of the laser pulse in nanoseconds. Range counter clock frequency determines the range measurement accuracy and is given by $\pm(c/2f_{\text{CLK}})$ where $c=3 \times 10^8$ m/s. A large number of manufacturers offer different modules of laser rangefinder electronics includ-

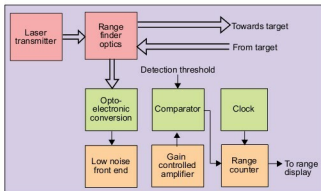


Fig. 10: Block schematic of different building blocks of receiver electronics of a typical laser rangefinder



Fig. 11: Range receiver electronics

ing photodetector amplifiers, fast pulse peak stretchers and range counters (Fig. 11).

Present-day laser rangefinders and target designators are largely configured around diode-pumped pulsed solid-state lasers. In that case, the transmitter electronics is nothing but the drive and control circuitry required for laser diodes used to optically pump the gain medium. Laser diode drive and control electronics mainly include current source to drive the laser diodes

and temperature controller to operate the laser diodes at the desired temperature. Different building blocks of laser diode electronics are discussed in the following parts of the article.

To be continued

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